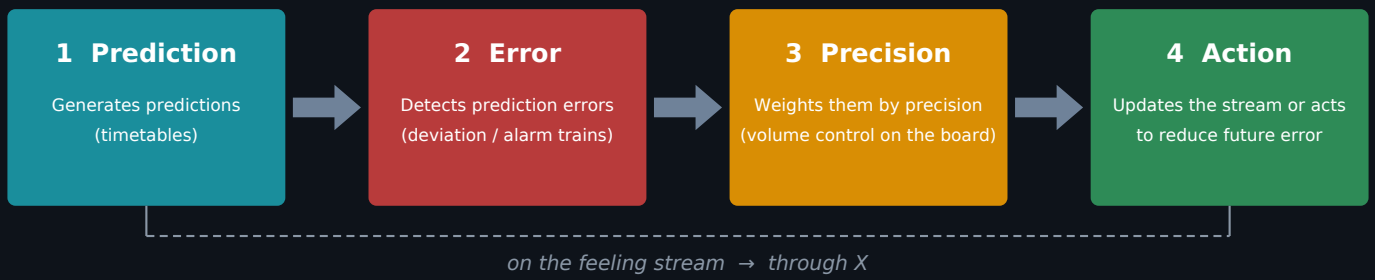
PSM – Network Explanation

The Railway Model of Mind · Draft · Working hypothesis · Not peer-reviewed

The mind runs as a continuous **feeling stream** — the main line. Everything that becomes conscious passes through a single now-point called **X** (the central station). The system constantly does four things:

1 Core idea



That is the whole engine: prediction, error, precision and action — running continuously on the feeling stream that passes through X.

2 The basic elements

Local stations

Specialised networks for raw data: senses, body (interoception), memory and language. Each station generates signals and can hold local predictions. Most traffic never needs the main platform.

Main line — feeling stream

The continuous subjective experience. Everything that passes through X colours or is coloured by this stream. Thoughts, words and actions ride on it; they do not replace it.

X — the now-point

The dynamic moment where the stream becomes experienced. X is not a physical place and not a processor. The stream does not stop at X; it passes through it.

The board — weight (w) + precision

Context, memories and priorities. Decides which trains get the platform and how strongly they colour the system. Precision is the volume knob: high amplifies, low dampens. Central to attention and emotional intensity.

Paint trains — emotions

Emotions colour the entire main line. Threat sprays anxiety. Joy is light. Anger is hot and fast. Sadness slows and dulls. Thickness of the paint is controlled by emotional (interoceptive) precision.

Action trains — Active Inference

The system can change the body or the world so predictions are fulfilled, instead of only updating the model. Reaching for a cup because you predict the sensation is an action train fulfilling a proprioceptive prediction.

3 The basic process

1	**Signal arrives**	Local station receives input
2	**Prediction running**	Top-down timetable is active
3	**Error calculated**	Deviation / alarm train generated
4	**Precision assigned**	Board sets the volume
5	**Access to X**	High precision → reaches the now-point
6	**Update or act**	Change model or send action train
7	**Stream continues**	Loop never stops

Steps 2–5 are **predictive coding**. Step 6 adds **Active Inference**. Everything runs on the feeling stream that passes through X.

4 How the main functions fit

Attention

Platform control + precision. The board raises precision on one train and lowers it on others. Only high-precision trains get the main platform at X. Distraction is a siding train suddenly receiving high precision.

Memory

Freight cars from archive stations. Strong emotional weight and high precision make them arrive faster and colour the stream more strongly. Traumatic memories carry extreme priority and can arrive uninvited.

Decision-making

The board simulates possible future routes (expected free energy) and gives green light to the action that best reduces future error while fulfilling preferred states. Impulsivity = a heavy temptation train forces through. Indecision = no route gets clearance.

Emotion

Paint trains colour the main line. Intensity is governed by emotional/interoceptive precision — the volume knob on bodily signals. High precision makes small bodily changes feel large. Low precision makes even strong signals thin and distant.

Language

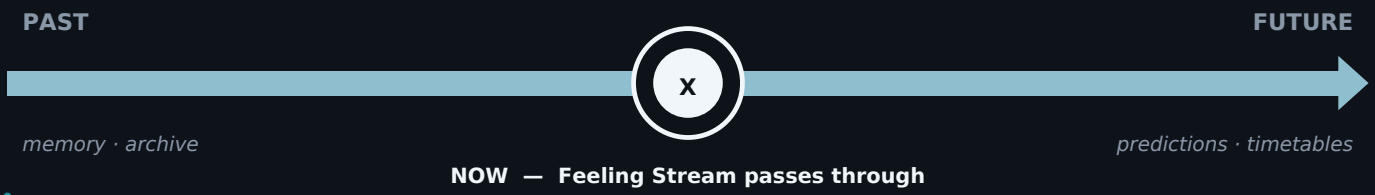
Language centres produce standardised container cars filled with words and symbols. These cars hook onto feeling and experience loads. Words give direction and labels, but they are not the stream itself. When you think in words, these symbolic cars roll through X, often coupled to emotional paint.

Sleep

At night the main line closes for maintenance. Work trains repair tracks, clear waste (glymphatic clearance), and move the day's experiences into the permanent archive. Dreams can be understood as test runs. Poor sleep leaves tracks damaged and the archive incomplete.

The feeling stream and X

The main line is primary. Predictive coding and Active Inference operate on it continuously: predictions colour it, prediction errors update it, actions try to steer it, and precision decides how much each signal may dominate. X is only the present point in this ongoing process.



The feeling stream is not produced by X. The stream passes through X. That distinction matters.

5 When the network fails

Psychological difficulties can be described as concrete faults in the network rather than as a 'broken mind'.

Anxiety & panic — locked loops + high precision

A threat paint train has permanently high precision and runs in a circle. Every lap reinforces the prediction of danger. Avoidance reduces short-term error but keeps the loop alive. Panic = the speed in the loop becomes critically high.

Depression — low voltage + rigid negative predictions

Few trains run. Those that arrive are slow and dark. Positive signals carry too little precision to update the model. Expected free energy for action is high, so action looks pointless. The main platform often stands almost empty.

Trauma — permanently deformed tracks

An extreme past event has bent the tracks. Neutral signals that resemble the old event automatically trigger the old high-precision alarm and paint train. Healing requires building safer alternative routes that gradually receive higher priority.

Other patterns

Obsessive patterns: action used rigidly to minimise highly precise threat predictions.
Addiction: action gives strong short-term reduction in free energy while distorting the board's longer-term priorities.

6 Core principles

1. **The feeling stream is primary.** Everything else — thoughts, words, decisions, actions — rides on or modulates the continuous stream of experience.
2. **X is the now-point, not a machine.** The central station is the present moment where the stream is experienced. It does not create the stream; the stream passes through it.
3. **Prediction + error + precision govern traffic.** Timetables (predictions), deviation trains (prediction errors) and the precision switch determine what reaches X and how strongly it colours the stream.
4. **Action is also a way of reducing error.** Active Inference: the system can change the body or the world so that predictions are fulfilled, instead of only updating the internal model.
5. **Emotional precision controls intensity.** The volume knob on paint trains (interoceptive precision) decides how strongly feelings dominate the stream and the decisions that follow.
6. **Failures are track faults, not a broken mind.** Anxiety, depression and trauma can be understood as locked loops, low voltage or deformed tracks — concrete network problems that can in principle be influenced.
7. **Language is hooks, not the origin.** Words are standardised cars that couple to feeling and experience loads. They give direction and labels, but they are not the stream itself.

7 Network overview

Local stations	Senses · Body · Memory · Language
Timetables	Predictions (top-down)
Deviation trains	Prediction errors (bottom-up)
Precision switch	Volume control on the board
Main line	Feeling stream
X	Now-point where the stream is experienced
Paint trains	Emotions (intensity = emotional precision)
Action trains	Active Inference — change body or world
Archive stations	Memory storage and retrieval
Symbolic cars	Language as hooks on experience

8 Summary

The railway model offers a single language for perception, emotion, decision-making and action. The feeling stream is the main line. X is the present point where that stream is experienced. Predictions run as timetables. Prediction errors arrive as deviation trains. Precision is the volume control on the board. Emotions are paint trains whose intensity is set by emotional precision. Action trains change the world so that predictions can be fulfilled.

When the network works well, predictions are updated, precision is flexible, and action reduces future error. When it fails, we see locked high-precision loops (**anxiety**), low voltage and rigid negative predictions (**depression**), or permanently deformed tracks (**trauma**).

The model is a working hypothesis. It is meant as a clear language for understanding and communication — not as a finished neurobiological theory. Its value lies in making the continuous interplay of prediction, precision, feeling and action visible in one coherent picture.